

# Efrat Dener, Ph.D. — Curriculum Vitae

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Plant ecology · seed dispersal & dispersal evolution · intraspecific trait variation · plant–pollinator interactions · statistical & computational ecology  
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269 citations · h-index 6 · i10-index 5 (Google Scholar, 2026). Journal impact factors: most recent JCR (2024).

## Research Focus

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I study how ecological and evolutionary forces shape variation in plant traits — especially seed dispersal — across organizational and spatial scales. A recurring role in my work is that of the integrative quantitative analyst on large, multi-year projects: I synthesize data generated by multiple students across distinct sub-studies into unified hierarchical, cross-scale analyses, supported by custom analytics dashboards. This integrative approach spans the multi-chapter dispersal-trait project with PhD student M. García Serrano, the Asiatic wild-ass ecology project, and my own doctoral research.

## Education

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Ph.D., Desert Studies / Ecology — Ben-Gurion University (BGU) 2016–2021 Albert Katz School for Desert Studies, Mitrani Dept. of Desert Ecology. Thesis: The interactive effects of genetic relatedness and selection pressures on seed dispersal distances in an annual plant. Supervisors: Prof. Itamar Giladi & Prof. Ofer Ovadia.

M.Sc., Desert Ecology — Cum Laude — BGU 2014–2016 Thesis: The effect of pollen source on dispersal-related traits in a heterocarpic annual plant. Supervisors: Dr. Itamar Giladi & Dr. Hagai Shemesh.

B.Sc., Environmental Sciences — Tel-Hai Academic College 2010–2014 Research project: The effect of temporally variable nutrient regimes on habitat selection in an annual plant. Supervisor: Dr. Hagai Shemesh.

## Academic Appointments

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Postdoctoral Researcher — Ben-Gurion University 2022–present Spatial Plant Ecology Lab (Prof. Itamar Giladi) — intraspecific trait variation, dispersal ecology, hierarchical & variance-partitioning modeling. Plant Communities & Complex Systems Lab (Prof. Merav Seifan) — plant–pollinator interactions, drought effects on plant fitness.

Postdoctoral Researcher — Tel-Aviv University 2021 Lab of Prof. Itay Mayrose — effects of genomic duplication (polyploidy) on plant–pollinator interactions and ecological-network structure. Collaboration with Prof. Tia-Lynn Ashman (Univ. of Pittsburgh).

## Publications

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Author legend: Dener, E. = author; † = equal/co-first contribution. Reverse chronological.

### Peer-reviewed journal articles

12. García, M.J., Dener, E. & Giladi, I. (2026). Multi-level variation in dispersal traits along a precipitation gradient. *Global Ecology and Conservation*, e04246. IF 3.7
11. Paul, S., Avidan, Y., Shahar, N., Dener, E., Bar-David, S. & Seifan, M. (2026). Dietary insights into ecological integration and impact on vegetation of a reintroduced large herbivore in a hyper-arid ecosystem. *Global Ecology and Conservation*, e04215. IF 3.7
10. Salman, I.N.A., Dener, E., Tzin, V. & Seifan, M. (2025). Plant fitness is shaped by cascading effects of aridity and drought on floral traits and pollination services. *Agriculture, Ecosystems & Environment* 393, 109855. IF 6.4 · 1 cit.
9. Krintza, N., Dener, E. & Seifan, M. (2024). Stress induces trait variability across multiple spatial scales in the arid annual plant *Anastatica hierochuntica*. *Plants* 13(2), 256. IF 4.1 · 7 cit.
8. Shemesh, H., Dener, E. & Sadeh, A. (2022). Bedrock may dictate the distribution of the fire salamander in the southern border of its global range. *Israel Journal of Ecology and Evolution* 69(1–2), 5–9. IF 0.8 · 3 cit.
7. Pozner, E., Bar-On, P., Livne-Luzon, S., Moran, U., Tsamir-Rimon, M., Dener, E., et al. (2022). A hidden mechanism of forest loss under climate change. *Forest Ecology and Management* 506, 119966. IF 3.7 · 49 cit.
6. Greenbaum, G., Dener, E. & Giladi, I. (2022). Limits to the evolution of dispersal kernels under rapid fragmentation. *Journal of the Royal Society Interface* 19(188), 20210696. IF 3.5 · 1 cit.
5. Cna'ani, A.†, Dener, E.†, Ben-Zeev, E., Günther, J., Köllner, T.G., Tzin, V. & Seifan, M. (2021). Phylogeny and abiotic conditions shape the diel floral emission patterns of desert Brassicaceae species. *Plant, Cell & Environment* 44(8), 2656–2671. IF 6.3 · 10 cit. († joint first authors)
4. Dener, E., Shemesh, H. & Giladi, I. (2021). The effect of pollen source on seed traits and dispersability in the heterocarpic annual *Crepis sancta*. *Journal of Plant Ecology* 14(3), 425–437. IF 3.9 · 6 cit.
3. Dener, E., Ovadia, O., Shemesh, H., Altman, A., Chen, S.-C. & Giladi, I. (2021). Direct and indirect effects of fragmentation on seed dispersal traits in a fragmented agricultural landscape. *Agriculture, Ecosystems & Environment* 309, 107273. IF 6.4 · 30 cit.
2. Chen, S.-C., Dener, E., Altman, A., Chen, F. & Giladi, I. (2020). Effect of habitat fragmentation on seed dispersal ability of a wind-dispersed annual in an agroecosystem. *Agriculture, Ecosystems & Environment* 304, 107138. IF 6.4 · 16 cit.
1. Dener, E., Kacelnik, A. & Shemesh, H. (2016). Pea plants show risk sensitivity. *Current Biology* 26(13), 1763–1767. IF 7.5 · 146 cit. (Widely covered in international science media.)

### Preprints / manuscripts under review

- Livne-Luzon, S., Yaakobi, A., Yalin, D., Sade, D., Dener, E., et al. (2026). Tree–microbe–soil interactions affecting soil organic carbon fractions in Mediterranean forest soils. EGU sphere preprint (submitted to *Journal of Ecology*).
- Ibdah, M., Kedoshim, R., Nawade, B., Cohen, Y., Dener, E., Revuru, B. & Shaltiel-Harpaz, L. Pollination protects against herbivory: unintended consequences of pesticides and volatile signal manipulation. Submitted to *Journal of Applied Ecology* (2026).
- Ostreiher, R., Dener, E. & Heifetz, A. Time perception guides foraging decisions in free-living

Arabian babblers. *Animal Behavior and Cognition* (in revision, 2026).

#### Manuscripts in preparation

- Dener, E., García, M.J. & Giladi, I. Environmental effects on variation in dispersal traits across organizational levels. (First author.)
- Dener, E. & Giladi, I. Dispersal traits in urban vs. non-urban environments across Israeli cities.
- Avidan, Y., Bar-David, S., Altman, A., Dener, E. & Seifan, M. No long-term effects of reintroduced large herbivores on vegetation composition in an arid region. (joint species distribution modeling + variance partitioning.)

#### Research in the Media

Dener et al. (2016), “Pea plants show risk sensitivity,” *Current Biology* — covered by 20+ international outlets following Oxford and Cell Press releases, including *Science/AAAS*, *Newsweek*, *Christian Science Monitor*, *The Daily Beast*, *The Scientist*, *IFLScience*, *Modern Farmer*, *UPI*, *Phys.org*, *ScienceDaily*; German outlets *wissenschaft.de* and *scinexx.de*; and the Hebrew Davidson Institute (Weizmann). Discussed in a *Current Biology* Dispatch commentary, “Decision-Making: Are Plants More Rational than Animals?” (2016).

#### Research Funding & Grant Contributions

| Year | Grant                                                                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020 | ISF Grant 2684/20 (contributing author) — “Levels of intraspecific variation in dispersal traits...” (PI: Giladi). ~848,000 NIS / 4 yrs. Created proposal figures; written feedback across drafts. Funds co-supervised PhD project. |
| 2024 | ISF Grant 2886/25 (contributing collaborator) — “Dust devils as a mechanism of secondary seed dispersal in drylands” (PI: Giladi). Proposal design, graphics, methods.                                                              |
| 2014 | The Sol Leshin Grant — 87,000 NIS. Co-initiated, developed and co-wrote (with Giladi & Shemesh).                                                                                                                                    |

#### Awards & Honors

| Year      | Award                                                                           |
|-----------|---------------------------------------------------------------------------------|
| 2021      | Publication Prize ×2 — Albert Katz Int’l School for Desert Studies, BGU         |
| 2021      | Postdoctoral Matching Scholarship — Tel-Aviv University                         |
| 2018      | Second-Best Student Presentation Award — 31st GfÖ Conference                    |
| 2016–2020 | The Negev (Faran) Scholarship for Outstanding Doctoral Students — BGU           |
| 2016      | Best Student Presentation Award — 29th GfÖ Conference                           |
| 2016      | The Merav Ziv Award for Outstanding Achievement in Ecology (M.Sc.) — BGU        |
| 2016      | “Recognition of Excellence” Prize; Dean’s Award — Albert Katz Int’l School, BGU |
| 2015–2016 | Full M.Sc. Scholarship — Albert Katz School for Desert Studies, BGU             |

#### Conference Presentations

| Year | Presentation                                                                                                                             |
|------|------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | Multi-level variation in dispersal traits along a precipitation gradient — PopBio Annual Conference (Poster)                             |
| 2020 | Habitat fragmentation & seed dispersal ability — 4th NSO/Oikos, Reykjavik (Poster)                                                       |
| 2019 | Evolution of dispersal kernels under fragmentation — 4th Plant Ecology in Israel, Tel-Hai (Oral)                                         |
| 2018 | Dispersal in fragmented habitats — 3rd Plant Ecology in Israel (Oral); 31st GfÖ, Innsbruck (Oral)                                        |
| 2016 | Long-distance pollination & short-distance dispersal — 29th GfÖ, Czech Rep.; Sede Boqer Symposium; 1st Plant Ecology in Israel (Oral ×3) |
| 2015 | Pollen source & dispersal-related traits — ISEES, Hebrew Univ. (Oral)                                                                    |
| 2014 | Temporally variable nutrient regimes & habitat selection — ISEES, Bar-Ilan Univ. (Oral)                                                  |

## Teaching Experience

Lecturer (Instructor of Record) — Tel-Hai Academic College

| Year      | Course                                    |
|-----------|-------------------------------------------|
| 2020–2022 | Statistics for Biologists / Biostatistics |
| 2020–2021 | Ecological Tracking as a Research Tool    |
| 2021–2022 | Research Methods B                        |
| 2022      | First Seminar (co-taught)                 |

Teaching Assistant / Guest Lecturer — Ben-Gurion University

| Year         | Course                                                                     |
|--------------|----------------------------------------------------------------------------|
| 2022–present | Data Visualization module, Biostatistics course                            |
| 2019–2021    | The World of Plants                                                        |
| 2017–2019    | Biophysical Field Methods in the Namib Desert (Namibia; in English)        |
| 2020         | Biostatistics & Experimental Design, Albert Katz Int'l School (in English) |

## Mentoring & Student Supervision

| Years        | Student & role                                                                                                     |
|--------------|--------------------------------------------------------------------------------------------------------------------|
| 2022–present | Manuel J. García Serrano — Ph.D. — co-supervisor (with Giladi)                                                     |
| 2023–2026    | Ibrahim Salman — Ph.D. — statistical & manuscript mentor; co-author                                                |
| 2024–2026    | Hemanth A. Yoganan — M.Sc. — statistical analysis & supervision                                                    |
| 2021–2026    | Additional M.Sc., B.Sc. & seminar students — thesis mentoring & experimental-design advising (BGU; Univ. Tübingen) |

## Professional Service & Outreach

Statistical consulting & collaboration (2017–present). Provide study-design, mixed/hierarchical-model, SEM and data-analysis support to 20+ research groups across ecology, conservation and the life sciences

— including paid consulting for academic labs at BGU and for organizations such as the Jewish National Fund (KKL). Several of these collaborations have produced co-authored publications.

Field surveys. Annual monitoring of protected wildflowers — *Iris atropurpurea* (Royal Iris) and *Paeonia* (Forest Peony), Upper Galilee — organized by Dr. Hagai Shemesh (2018, 2021, 2023, 2025, 2026).

AI & computational tools in education. Interactive simulations of classical ecological models (Prof. Seifan’s Community Ecology course); ChatGPT-based botanical-identification tool (Prof. Giladi’s courses, 2025); contributing to AI integration in the new international B.Sc. in Renewable Natural Resources, Sde Boker (2026).

Committees & service. Organizing committee, SIDEER symposia, BGU (2018–2020); Academic Committee, Student Council, Sde Boker (2016–2020); session moderator, BGU student conference (2025); curriculum design, new B.Sc. in Bio-Data / Computational Biology, Tel-Hai (2021).

Public outreach. “Introduction to Science” with high-school students (2019–2020); ecology education with elementary-school children (2018).

## Technical & Computational Skills

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Statistical modeling. LM/GLM/LMM/GLMM; variance partitioning & hierarchical models and SEM / path analysis (specialties); bootstrap & resampling; phylogenetic comparative methods (PGLS, phylogenetic path analysis); multivariate ordination (PCA, NMDS, cluster); spatial & time-series analysis; simulation / individual-based modeling; experimental design.

Programming & reproducibility. R / RStudio (advanced — Shiny, R Markdown / Quarto); Python (data pipelines); NetLogo; Matlab; Git. SPSS, JMP/Jamovi, AMOS, Tableau.

GIS & imaging. QGIS, ArcMap; ImageJ.

Field & laboratory. Designed and ran large-scale common-garden (net-house) experiments and multi-year field seed-collection campaigns (heterocarpic annuals; nested sampling across populations, mother plants, fruits and seeds). Multi-year arid/Mediterranean vegetation surveys; controlled pollination & wind-dispersal experiments; molecular-biology & analytic-chemistry methods.

Research software development. Multiple multi-module R Shiny dashboards for integrating, exploring and visualizing large multi-source datasets (each with an automated data pipeline + test suite); Python data-processing pipeline with API integration; iOS (Swift) app. End-to-end research-software capability: reproducible workflows, automated pipelines, interactive analytics, version control.

## Additional Training

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| Year | Activity                                                                                       |
|------|------------------------------------------------------------------------------------------------|
| 2018 | Research visit (~1 month), lab of Dr. P.-O. Cheptou, CEFÉ Montpellier, France                  |
| 2017 | Agent-Based Modelling with NetLogo (Univ. of Göttingen); Quantitative Methods in Ecology (BGU) |
| 2015 | Biostatistics: ANOVA & Design of Experiments (BGU)                                             |

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## Languages

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Hebrew (native) · English (near-native, C2).

## References

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Available upon request.